

LEO FRANKEL

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EDUCATION

The University of Chicago | Chicago, IL

Expected June 2027

Bachelor of Science, Mathematics

GPA: 3.83/4.0

Bachelor of Science, Computer Science (Theory Track)

Coursework: Abstract Linear Algebra, Honors Discrete Mathematics, Real Analysis, Honors Algorithms

Oakwood Secondary School | North Hollywood, CA

August 2017 – June 2023

Cumulative GPA: 4.0 / 4.0 (unweighted)

National Merit Scholarship Recipient

RESEARCH

WEFT Programming Language | Remote

March 2025 – Present

Designer + Developer [\[demo\]](#)

- Design WEFT, a novel reactive programming language implementing unified signal abstraction for real-time multimedia composition, enabling media-agnostic cross-domain creative work.
- Develop complete language specification including evaluation semantics, type system, and runtime architecture for declarative time-based media programming

Programming Language Design Seminar | Remote

June 2025 - Present

Independent Study with Prof. Alessandro Warth and John Underkoffler

- Build multiple programming language prototypes, exploring diverse paradigms and implementation approaches under mentorship of language design researchers
- Study advanced parsing techniques, type systems, and runtime architectures through hands-on language development projects

University of Chicago Dept. of Mathematics | Chicago, IL

June 2024 – August 2024

Apprentice Research Program

- Authored 17-page introductory paper on modular forms, exploring their importance, fundamental properties, and dimension formulas
- Created original mathematical visualizations to illustrate concepts including fundamental domains and modular symmetries

WORK

Redflag AI | Remote

June 2025 – September 2025

Software Engineer

- Designed TVIP conditional access system for Televisa that won \$5M contract, implementing comprehensive content delivery and protection architecture
- Built real-time piracy detection and takedown systems for live sports and entertainment content with sub-10-second response times

Doralice | Remote

November 2024 – Present

Assistant Creative Director

- Develop design systems for Apple Music, creating visual identities and brand guidelines for product launches and apparel.
- Support creative problem-solving for high-profile client projects, including studio launch campaigns and multi-platform brand rollouts.

LEADERSHIP

Boy Scouts of America, Troop 223 | Los Angeles, CA

April 2016 – May 2021

Eagle Scout

- Led 250+ scouts as Senior Patrol Leader, organizing weekly troop meetings, coordinating camping trips, and delegating responsibilities to patrol leaders
- Designed and constructed outdoor storage facilities for a local homeless shelter, managing a 6-month project including budget planning, volunteer coordination, and construction